

A Packing of 8 Four-Spaces in $\mathbb{R}^{12}$

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Abstract

We give 8 4-dimensional subspaces of $\mathbb{R}^{12}$ whose smallest squared chordal distance is at least 3.047365. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.999732, obtained by deleting 1 of the 9 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 8 subspaces of dimension 4 in $\mathbb{R}^{12}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 3.047365\dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 12$, $n = 4$. The cell $N = 8$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 2.999732, obtained by deleting 1 of the 9 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(12 - 4)/12 \cdot 8/7 = 3.047619$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m12n4N8.txt`

in <https://github.com/Ge-limin/math>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 61 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
304736597168625614835044166683086384450455888960176957948701619620576941
789948836499438352368258508300838983530539282972508283415015024270610905
8560160572892605999193406030709223909315327065755
```

denominator:

```
100000000000268500755289856637847273539509356587363269327508507341415344
737101563931468880943190926055637145389571694855292306803944825784335187
7541424814785758077645510478041904262767696986568
```

B Coordinates

8 blocks of 12 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m12n4N8.txt`.

```
473293940850 59937715366 -24827542985 69281635692 125501928214 52897697818 375509587131 464005898301 195317321885 289643774591 -69965937633 -228392072396
-80301945120 318318216968 288400632403 111675526888 389078532576 -187577410232 310037780821 -257812804040 56916522636 -176396469175 769969768356 -213352893321
8132684200 612509175390 -301174069182 -204258831957 132683278848 -118484924450 -577555439337 -576416949770 10549094 431285457822 274648669876 227341707984
-405488340930 -150238545744 230017353217 -383705479305 84250978 -28400311567 206206142532 65213564165 -312900045961 -81667392516 -73811520400 490921776226
-124948138980 468117830270 452713504385 368913746793 5240952479 -336232347358 266900252997 -577460756128 774582393 -482745885144 -3075041610 -176296530640
-126024402300 -59105483369 -240402694413 456858252403 318976500 -175878300897 -68161469516 -35960946014 46596853501 242553344259 -260010729550 -309072620373
217577115231 -39278568918 -507628089061 49920412245 100129789935 -255524782974 -259127456924 -96193804040 8051445093 -167572936251 -101464836125 -127399577982
208504207026 194016055303 -63569553849 33066992552 -381683338281 6049935879 -148084245494 49449599214 434509449761 -224126819449 -295163983442 -340573030418
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-298347042951 289247182842 196827590195 4909362445 78660487674 -271071737746 -333798170844 -234584957120 3615296135 -119516749393 -197972957474 429338770761
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413276777033 366120383586 -124054077080 27225789081 98165700933 99096163400 265141871541 -51615556814 -538131643369 600128901248 116125883154 -168053465543
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-103683181091 -251380175751 554864719144 26209735103 8172334402 -451503582316 207945195846 57297429602 20290149244 -398037728083 -348803692005 131037869567
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